

# Intermediate Docker Commands

INTERMEDIATE DOCKER



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# Docker refresher

- Docker is a container runtime
- Is designed to run and manage various containerized applications on Windows, Mac, and Linux
- Can run containers using pre-built images, or create our own
- Dockerfiles are text files used to Docker container images
- Containers are instances of a given Docker image

The DataCamp [Introduction to Docker course](#) is a pre-requisite to this course

A screenshot of a course card for 'Introduction to Docker' on DataCamp. The card has a dark blue background. At the top, it says 'INTERACTIVE COURSE' in small white letters. Below that, the title 'Introduction to Docker' is in large white letters. There are two buttons: a green 'Continue Course' button and a white 'Bookmark' button with a yellow bookmark icon. At the bottom, there are four tags: 'Intermediate' with two green dots, '4 hours' with a clock icon, '15 videos' with a play icon, and '56 exercises' with a code icon.

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## Introduction to Docker

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# Docker commands

- `docker run`
- `docker stop`
- `docker build`
- `docker --help`
  - Provides a list of potential Docker commands
- `docker COMMAND --help`
  - `docker run --help`
    - Provides options for the `docker run` command

# Temporary containers

- Docker containers are usually created with `docker run`
- Containers remain even after stopping / exiting
- Often want to run a container instance and remove it immediately upon exit
  - Development
  - Testing
  - Scripts
- `docker run --rm`
  - `docker run --rm alpine:latest`
    - `/bin/sh`
- Referenced as 'clean-up' or 'remove'

# docker ps

- Used for determining name, id, status, and other attributes of containers on a given machine running Docker
- Use the `-a` flag to get more information about existing containers
  - `docker ps -a`
- Will cover how to get extremely detailed information about containers later in the course

```
$ docker ps -a
```

| CONTAINER ID | IMAGE  | COMMAND                 | CREATED            | STATUS                        |
|--------------|--------|-------------------------|--------------------|-------------------------------|
| 588bb3a3134e | nginx  | "/docker-entrypoint..." | 49 seconds ago     | Up 47 seconds                 |
| 8a0578a9b33c | ubuntu | "/bin/bash"             | About a minute ago | Exited (0) About a minute ago |

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# Mounting the host filesystem

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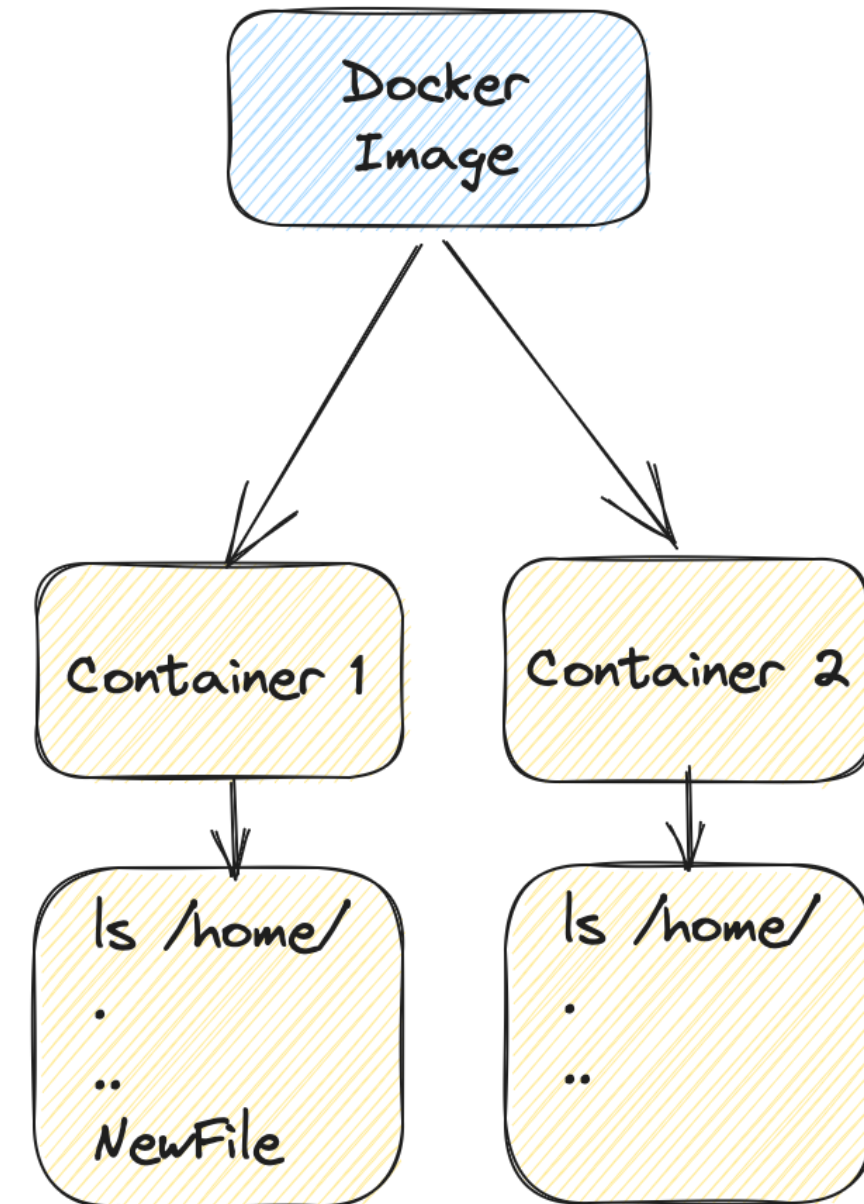


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# Container filesystems

- Container instances each have their own filesystem
  - Based off the image the container was created with
- Any changes are tied to that specific container instance
- Any changes are maintained across restarts
  - For that instance **only**
- New containers only have the data in the image, not instance specific changes





# Sharing files or directories

- Can attach specific files or directories to containers
- Allows for persistence of data, without maintaining a specific container
- Can upgrade container to new version but safely keep data / changes
- Known as **bind-mount**
- Can be read-only or read/write
- Note: When files or directories are attached to a container, they are not accessible to the host until the container is shutdown

# Using the -v option

- bind-mounts most often use the `-v` flag
- `-v <source>:<destination>`
- Multiple `-v` commands permitted
- Can also use the `--mount` option
- Note: bind-mount **hides** any content already present in the destination directory

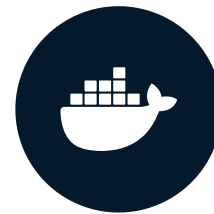
```
docker run -v ~/html:/var/www/html \  
    nginx
```

```
docker run  
    -v ~/pgdata:/opt/data \  
    -v ~/pg.conf:/etc/pg.conf \  
    postgresql
```

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# Persistent volumes

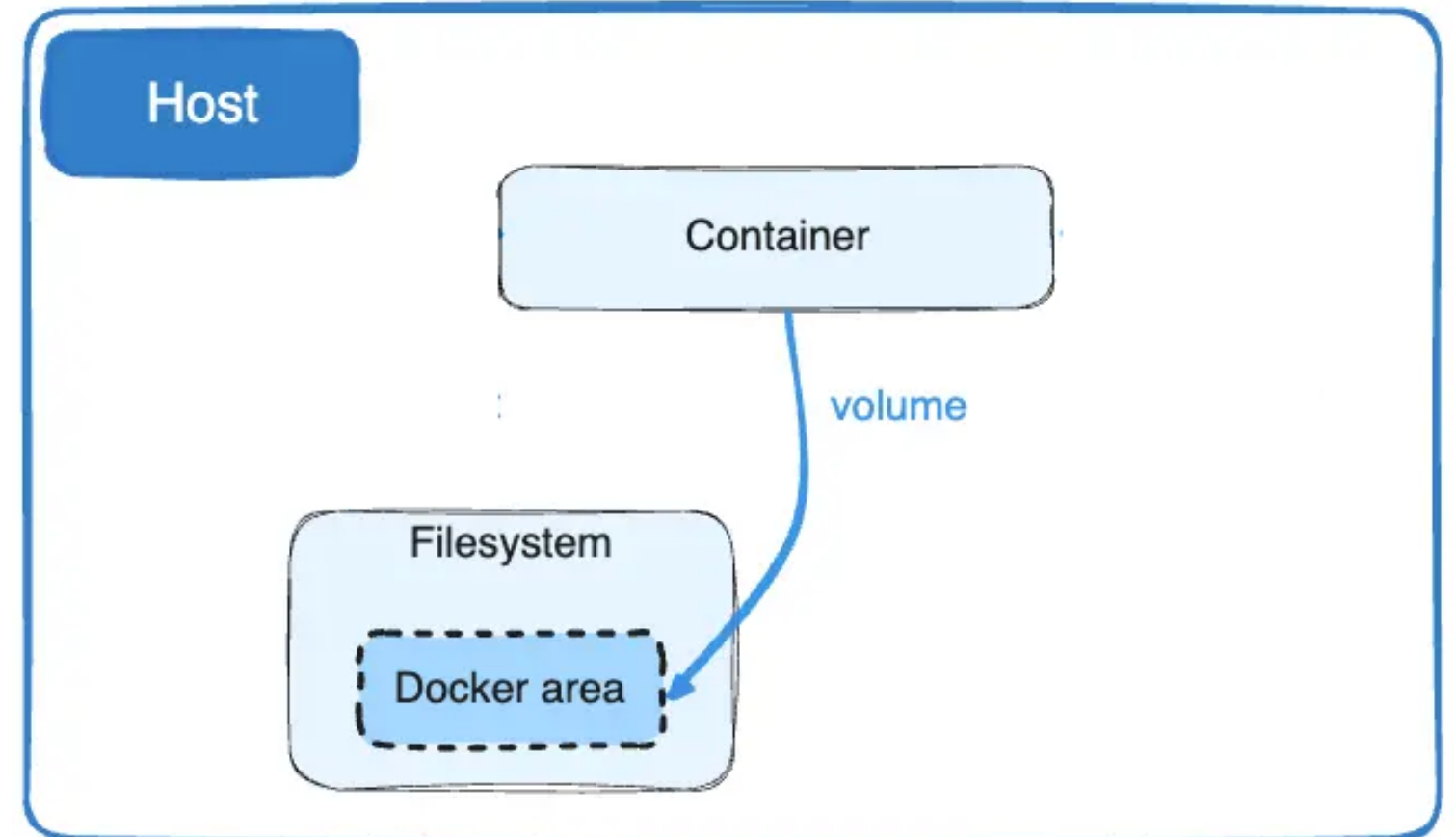
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# What is a volume?

- **Volumes** are an option to store data in Docker, unrelated to the container image or host filesystem
- Are managed from the command line (or API)
- Can share with multiple containers
- Higher performance than file share / bind mounts
- Exist until removed



<sup>1</sup> Image modified from <https://docs.docker.com>

# Managing volumes

- `docker volume`
- `docker volume create <volumename>`
- `docker volume ls` or  
`docker volume list`
- `docker volume inspect`
  - Provides assorted metadata about the volume, including Name, Mountpoint, Options, and so forth
- `docker volume rm`

# Volume creation example

```
bash> docker volume create sqldata
```

```
sqldata
```

```
bash> docker volume ls
```

| DRIVER | VOLUME NAME                                                      |
|--------|------------------------------------------------------------------|
| local  | 2f2b7f710551e004dcdd9edf4cad31c37826b428de12f1c04ca02305d216ab00 |
| local  | 14da7ff0c6eb29f644e6f9f9d59bbcf56b3699c04881dd7cbcaa9ecd6bef239c |
| local  | 150aa3c5c7aee30ffd1ec7ecf39f03989bf561536a9413ebed96ffbaa537d103 |
| local  | sqldata                                                          |
| ...    |                                                                  |



# Volume inspect example

```
bash> volume inspect sqldata
[
  {
    "CreatedAt": "2024-01-27T04:27:51Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/sqldata/_data",
    "Name": "sqldata",
    "Options": null,
    "Scope": "local"
  }
]
```



# Attaching volumes

- Uses the `-v` command
  - `docker run -v <volumename>:`
    - `<destination path>:<options>`
    - Volume name is name of existing volume
    - Destination path is the location the volume will be mounted (such as `/data`)
    - Options are optional comma-separated list of values such as `ro` for read-only.
- `--mount` exists as with bind-mounts

```
$ docker run -v sqldata:/data postgres
```

# Drivers

- Methods of storing Docker volumes
- Can include:
  - Local filesystem (default)
  - NFS (Unix filesharing)
  - SMB / CIFS (Windows filesharing)

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